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A Flexible Pressure Sensor with Wide Range Using Polyimide/Graphene Oxide/$BaTiO_3$ Nanofibers as Dielectric Layer *

Dezhi Wu, Xianshu Cheng, Yunheng Wu, Zhenjin Xu*

***Abstract*—Flexible pressure sensors capable of sensing wide pressure range have been attracting extensive attention from both academia and industry due to its versatile applications, such as aircraft skin and intelligent robots etc. Herein, an electrospun Polyimide/Graphene Oxide/$BaTiO_3$ nanofiber membrane was used as a dielectric layer sandwiched between two PI membranes with copper electrodes to fabricate a flexible capacitive pressure sensor with wide working range. The fabrication process, tactile perception mechanism and sensing performance of the sensor were investigated. Experimental results show that the pressure sensor exhibits superior sensing range (1.15 MPa), short response/recovery time (39 ms / 26 ms) as well as good stability (>2000 cycles under the pressure of 200 kPa). Application demonstration for foot motion monitoring indicates that such method will be one of prospective candidates to monitor pressures in many advanced applications in the future.**

Keywords-flexible capacitive pressure sensor, composite nanofiber, wide pressure range, tactile sensor

I. INTRODUCTION

In recent years, flexible pressure sensors with high sensitivity and wide sensing range have received a lot of attentions[1, 2] owing to the increasing demands in the fields of advanced applications, such as human-machine interations[3, 4] and health monitoring[5, 6]. Various mechanisms have been extensively proposed to achieve high sensing performance such as capacitance[7], piezoelectricity[8] and piezoresistance[9]. Among them, capacitive pressure sensors have been investigated intensively because of their good stability, simple structures and easy manufacturing processes.

The high sensitivity over a wide pressure range is very important for the practical applications. In past decade, a lot of methods have been carried out to improve the sensitivity and working range of the flexible sensors. For example, Prubost et al. fabricated a flexible pressure sensor based on a carbon black / PDMS syntactic foam film, which had the sensitivity of 35 kPa^{-1} (<0.3 kPa) and the working range of 0~12 kPa [10]. Lei et al. proposed a carbon nanotubes-coated polydimethylsiloxane foam to reach a maximum sensitivity of 67.02 kPa^{-1} over a large range between 25~270 kPa[11]. Although these reported flexible pressure sensors have high sensitivity or large working range, it is still a critical remaining challenge in achieving ultra-large sensing range up to MPa, which limits the potential applications in high-pressure conditions, such as aircraft skin and gait recognition. Due to the high porosity, PI nanofiber membranes typically possess extremely low bending stiffness and significant resistance to mechanical bending, showing great advantages in flexible electronics [12, 13]. But the sensor using polyimide (PI) nanofiber membrane as dielectric layer has low sensitivity[14], resulting in the limited applications. It can be seen from recent papers that functional nanomaterials[7, 15] and ceramic nanomaterials[16, 17]are helpful to improve the sensitivity of sensors. Therefore, it is reasonable to assume that the relationship between sensitivity and wide sensing range can be well balanced via adding functional nanomaterials and ceramic nanomaterials to PI nanofiber.

Here, we propose a flexible capacitive pressure sensor for wide-range pressure monitoring. Porous electrospun PI / GO / $BaTiO_3$ nanofiber membrane is utilized as a dielectric material to allow large deformations. Due to the high porosity of PI / GO / $BaTiO_3$ nanofiber membrane, the flexible pressure sensor exhibits high sensitivity of 11.4 MPa^{-1} for the range of 0~323 kPa, 6 MPa^{-1} for the range of 323~603 kPa and 3.37 MPa^{-1} for the range of 603~1152 kPa. Furthermore, a short response time of 39 ms, fast recovery time of 26 ms, good repeatability of 2000 cycles under 200 kPa have been measured. As a demonstration example, the sensor has been attached to the insole for gait recognition application.

II. EXPERIMENTAL

A. Electrospinning of PI / GO / $BaTiO_3$ nanofiber membrane

The preparation procedure of electrospinning precursor solution is as follows: Firstly, 0.0885 wt% GO powder purchased from Sigma-Aldrich is dissolved in DMAC solution and put into the ultrasonic system (KUDOS, Shanghai) at 53 kHz for 2h. Afterwards, $BaTiO_3$ ceramic nanoparticles is dispersed in the matrix prepared in the previous step by ultrasound. In the next part, Pyromellitic dianhydride (PMDA) and 4,4′-Diaminodiphenylether (ODA) purchased from Aladdin with molecular weight of 218.12 g / mol and 200.24 g / mol respectively are added and magnetically stirred (IKAC MAG HS 7, Germany) at room temperature for 12 h, forming the 30wt% precursor solution. Then, it is pumped by a precision pump (Harvard 11 Pico Plus) to a flat needle with the diameter of 250 μm. The PAA / GO / $BaTiO_3$ nanofiber membrane is electrospun with high DC voltage and the parameters used during the process is listed in

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Table 1. Finally, the fabricated PAA / GO / $BaTiO_3$ nanofiber membrane is placed in a vacuum furnace (VBF-1200X, China), kept at 100 ºC for 1 h (solvent removal), at 250 ºC for 1 h (dehydration cyclization) and at 350 ºC for 1 h (thermal amination) to convert the PAA / GO / $BaTiO_3$ nanofiber membrane to PI / GO / $BaTiO_3$ nanofiber membrane.

TABLE I. PARAMETERS FOR ELECTROSPINNING OF PI /GO / $BaTiO_3$ NANOFIBER MEMBRANE

Parameters	Value
PI solution concentration	30 wt%
GO concentration	0.0885 wt%
$BaTiO_3$ concentration	10 wt%, 30wt%, 50wt%
Applied voltage	10 kV
Distance between nozzle and collector	15 cm
Feed rate	0.3 ml / h

B. *Package of the sensor*

The 50 μm PI / GO / BaTiO3 nanofiber membrane is sandwiched between two PI membranes with 6×6 mm copper electrode (Fig 1a). Finally, the sensor is packaged with glue and attached to a prosthetic hand made of wood, as shown in Fig 1b.

C. *Characterization and measurements*

Morphologies of electrospun PI / GO / $BaTiO_3$ nanofiber membrane are characterized by using scanning electron microscopy (SEM, LEO 1530). Digital dynamometer (HP-100, Aidebao, China) with a range of 0-100 N is used to measure the applied force. The sensor is fixed on the sample platform with the electromechanical universal testing machine and connected to the LCR meter (Agilent E4980AL, Keysight technologies, USA) and computer to measure the corresponding capacitance change, as depicted in Fig 1c and Fig 1d.

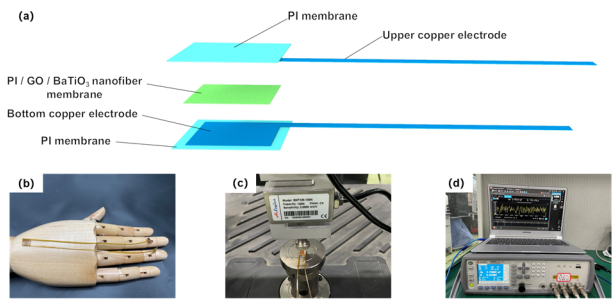

Figure 1. (a) schematic diagram of the flexible capacitive pressure sensor using PI / GO / $BaTiO_3$ nanofiber membrane as dielectric layer for wide-range pressure monitoring. (b) packaged pressure sensor on prosthetic hand. (c) pressure sensor on digital dynamometer for pressure testing. (d)LCR meter and computer for capacitance acquisition.

III. RESULTS AND DISCUSSIONS

A. *Fiber morphology*

As shown in Fig 2 the electrospun PI / GO / $BaTiO_3$ nanofiber membrane has good porosity and fiber diameter increases with increasing $BaTiO_3$ concentration. Supply voltage, distance and feed rate are the three most important parameters for electrospinning. It can be seen from the Fig 2d fiber diameter decreases with supply voltage. Analytically, the higher voltage can provide a larger electrostatic repulsion force for the jet, which is beneficial to stretch the jet, thereby reducing the fiber diameter. As shown in Fig 2e, it is observed that fiber diameter decreases with distance. As the distance increases, the jet is fully stretched, resulting in a reduction in fiber diameter. On the contrary, with the increase of the feed rate, the fiber diameter gradually decreases (Fig 2f). The reason for this phenomenon is that the high feed rate increases the diameter of jet. When the electric charge is the same, the splitting ability of the coarse jet is poor, so that the diameter of nanofibers is larger. Therefore, we can control the morphology and diameter of the fibers by adjusting the solution concentration, voltage, distance and feed rate.

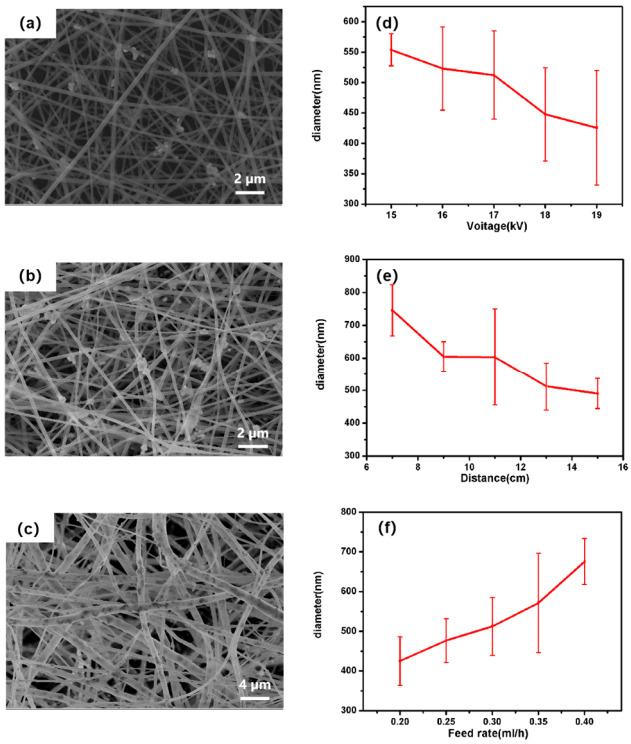

Figure 2. PI / GO / BaTiO3 nanofiber membrane morphology. (a) 10 wt% $BaTiO_3$. (b) 30 wt% $BaTiO_3$. (c) 50 wt% $BaTiO_3$. (d~f) the relationship between fiber diameter and voltage, distance and feed rate respectively.

B. *Sensing principle and performance*

As presented in the Equation (1~2), the absolute permittivity in vacuum (ε_0) is 8.86×10^{-12} F/m ; ε_r is the relative permittivity of the dielectric layer; A, d are the facing area and distance of the upper and lower electrode, respectively; ε_{air} = 1, V_{air} and $V_{nanofiber\ mat}$ refer to the volume ratio of air and PI / GO / $BaTiO_3$ nanofibers in the porous nanofiber dielectric layer, respectively.

$$\varepsilon_r = \varepsilon_{air} V_{air} + \varepsilon_{nanofiber\ mat} V_{nanofiber\ mat} \tag{1}$$

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d} \tag{2}$$

Once the pressure is applied to the sensor, the contacted part of PI / GO / $BaTiO_3$ nanofiber membrane is compressed, V_{air} and d gradually decrease, leading to an increase of ε_r and capacitance (C).

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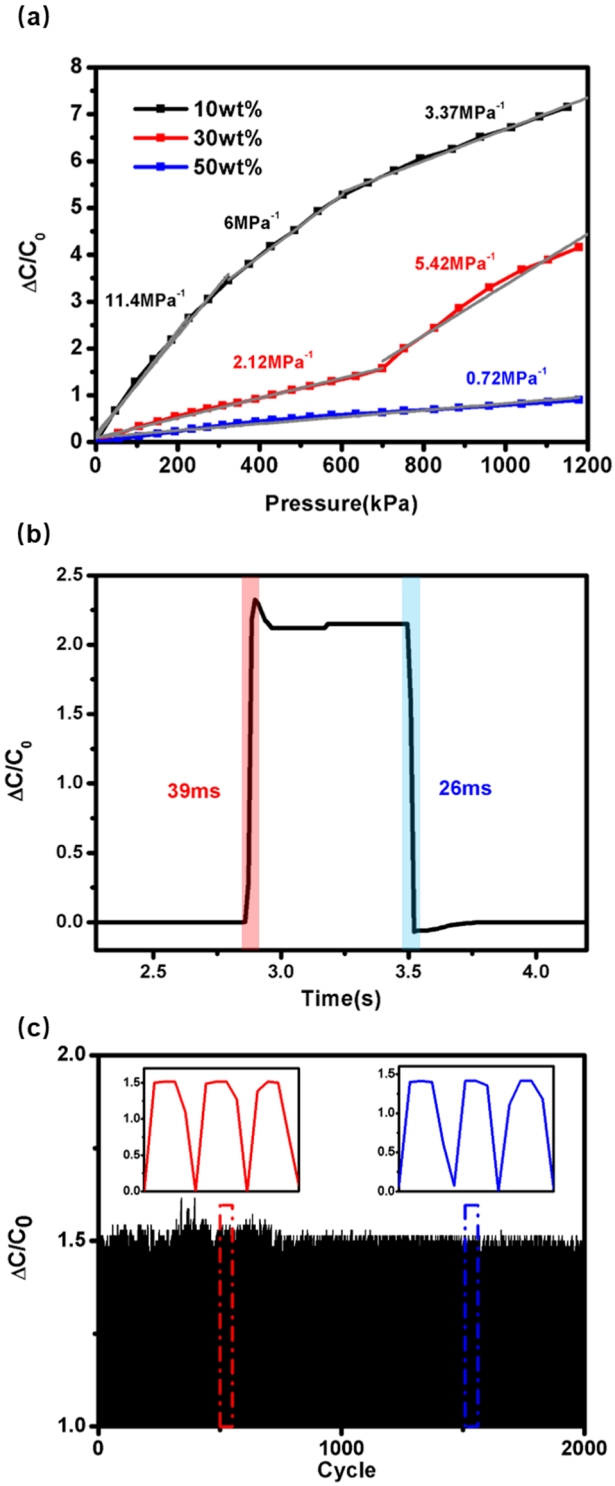

Figure 3. Sensing characteristics. (a) the sensitivities of the sensors with different mass ratio of $BaTiO_3$. (b) the capacitance response and recovery time of the sensor under a rapid loading of 280 kPa (c) the relative capacitive change of the sensor under repeated loading and unloading 200 kPa pressure.

The effect of the concentration of $BaTiO_3$ on fiber morphologies is also discussed. As shown in Fig 3a, we compared the effect of different concentrations of $BaTiO_3$ on the sensor performance. When the pressure is applied, the sensitivity of the sensor with 10 wt% $BaTiO_3$ is 11.4 MPa^{-1} in the pressure range of 0~323 kPa, 6 MPa^{-1} in the pressure range of 323~603 kPa and 3.37 MPa^{-1} in the pressure range of 603 ~1152 kPa. The sensitivity of the sensor with 30 wt% $BaTiO_3$ is 2.12 MPa^{-1} in the pressure range of 0~700 kPa and 5.42 MPa^{-1} in the pressure range of 700~1180 kPa. The sensitivity of the sensor with 50 wt% $BaTiO_3$ is 0.72 MPa^{-1} in the pressure range of 0~1180 kPa. Therefore, it can be concluded that the sensor with 10 wt% $BaTiO_3$ has the highest sensitivity in the ultra-large sensing range. Here, the dynamic response under a rapid pressure of 280 kPa has been measured in Figure 3b, showing a response time of 39 ms and a recovery time of 26ms. Figure 3c shows the good repeatability after 2000 cycles under the pressure of 200 kPa. The high sensitivity in the ultra-large sensing range, fast dynamic response and good repeatability of the sensor demonstrates the considerable application potential in vehicle safety systems and gait recognition.

C. Application of the capacitive flexible pressure sensor for gait recognition

With superior performance, the flexible pressure sensor is used to test the various motions of feet. The function and structure of the human feet largely meet the needs of the feet movements. Once there is a problem with the feet, people's daily life will be affected to a large extent. In recent years, studies have shown that the plantar pressure during walking is an important basis for the diagnosis of abnormal foot morphology, and high plantar pressure is an important risk factor for foot development[18]. As shown in Fig 4, the sensor is attached to the insole of a male weighing 60 kg for monitoring feet movements. At the beginning of the measurement, the tester puts on shoes and keeps standing, the capacitance will have a significant change of 6 times and remain unchanged for a period of time. Then the tester starts to walk, the capacitance change is similar to that of standing, and shows a regular increase and descending. Finally, the tester starts running, the capacitive change at this time is higher than the standing and walking, since running will bring some impacts on feet. In a word, outstanding experimental results demonstrate that the capacitive pressure sensor provide a facile, effective method to monitor feet movements and promote the feet diagnosis owing to its good stability, small size, high sensitivity, high flexibility and ultra-large sensing range.

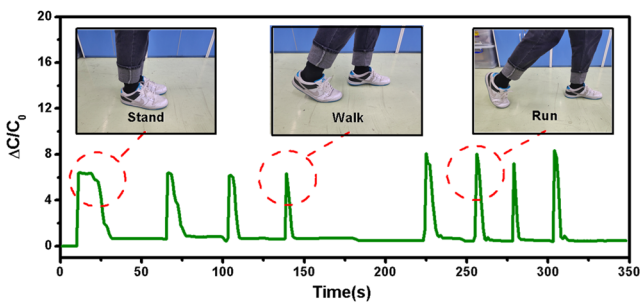

Figure 4. Application of the flexible pressure sensor for motion monitoring.

IV. CONCLUSION

In this work, we used an electrospun PI / GO / $BaTiO_3$ nanofiber membrane as dielectric layer for a capacitive pressure sensor with high sensitivity in ultra-wide sensing range. The sensor demonstrates high sensitivity of 11.4 MPa^{-1} (0 ~ 323 kPa), ultra-wide working range of 1.15 MPa, short response/recovery time, outstanding stability. Experimental

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results demonstrate that PI / GO / $BaTiO_3$ with the mass fractions of 10% can greatly enhance the sensing performance. The capacitive flexible pressure sensor has been successfully applied in gait recognitions, indicating that such method will pave a new way for fabricating flexible pressure sensors with large working range.

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